

Practical Workbook

CT-261

DATABASE MANAGEMENT SYSTEMS



Name: **Moazzam Farooqui**

Roll No: **CT-24068**

Batch: **2024**

Department: **CSIT**

Practical Workbook

Database Management Systems

(CT - 261)

Prepared by
Dr. Umer Farooq
CS & IT

Approved by
Chairman

Department of Computer Science & Information Technology NED University of
Engineering & Technology

Table of Contents

S. No.	Object	Date	Signature
01	Basic SQL SELECT Statements	14-JAN-26	
02	Restricting and Sorting Data	14-JAN-26	
03	Single-Row Functions	21-JAN-26	
04	Displaying Data from Multiple Tables	21-JAN-26	
05	Aggregating Data Using Group Functions	28-JAN-26	
06	Using Sub queries	04-FEB-26	
07	Manipulating Data	18-FEB-26	
08	Creating and Managing Tables	13-APR-26	
09	Constraints	13-APR-26	
10	Creating Views	20-APR-26	
11	PL/SQL Declaring Variables	26-APR-26	
12	PL/SQL Writing Executable Statements	05-MAY-26	

Q1. Evaluate the PL/SQL block provided in the text and determine the data type and value of each of the following variables according to the rules of scoping:

- a) The value of V_WEIGHT at position 1 is:
- b) The value of V_NEW_LOCN at position 1 is:
- c) The value of V_WEIGHT at position 2 is:
- d) The value of V_MESSAGE at position 2 is:
- e) The value of V_NEW_LOCN at position 2 is:

SOLUTION:

- a) V_WEIGHT at position 1 (inner block): Value: 2, Data Type: NUMBER. (Initialized to 1, then incremented by 1).
- b) V_NEW_LOCN at position 1 (inner block): Value: 'Western Europe', Data Type: VARCHAR2. (Initialized to 'Europe', concatenated with 'Western ').
- c) V_WEIGHT at position 2 (outer block): Value: 601, Data Type: NUMBER. (Outer variable initialized to 600, then incremented by 1).
- d) V_MESSAGE at position 2 (outer block): Value: 'Product 10012 is in stock', Data Type: VARCHAR2. (Initialized to 'Product 10012', concatenated with ' is in stock').
- e) V_NEW_LOCN at position 2 (outer block): Out of scope / Illegal. This variable is declared exclusively in the inner block and cannot be accessed or assigned to in the main block.

Q2. Suppose you embed a subblock within a block, as shown in the text. You declare two variables, V_CUSTOMER and V_CREDIT_RATING, in the main block. You also declare two variables, V_CUSTOMER and V_NAME, in the subblock. Determine the values and data types for each of the following cases:

- a) The value of V_CUSTOMER in the subblock is:
- b) The value of V_NAME in the subblock is:
- c) The value of V_CREDIT_RATING in the subblock is:
- d) The value of V_CUSTOMER in the main block is:
- e) The value of V_NAME in the main block is
- f) The value of V_CREDIT_RATING in the main block is

SOLUTION:

- a) V_CUSTOMER in the subblock is: Value: 201, Data Type: NUMBER(7). (Local subblock declaration overrides main block).
- b) V_NAME in the subblock is: Value: 'Unisports', Data Type: VARCHAR2(25).
- c) V_CREDIT_RATING in the subblock is: Value: 'EXCELLENT', Data Type: VARCHAR2(50). (Inherited from the main block as it is not redeclared locally).
- d) V_CUSTOMER in the main block is: Value: 'Womansport', Data Type: VARCHAR2(50).
- e) V_NAME in the main block is: Out of scope. (It was declared only within the subblock.
- f) V_CREDIT_RATING in the main block is: Value: 'EXCELLENT', Data Type: VARCHAR2(50).

Q3. Create and execute a PL/SQL block that accepts two numbers through iSQL*Plus substitution variables.

- a) Use the DEFINE command to provide the two values. (e.g., DEFINE p_num1 = 2 and DEFINE p_num2 = 4)
- b) Pass the two values defined in step a above, to the PL/SQL block through iSQL*Plus substitution variables. The first number should be divided by the second number and have the second number added to the result. The result should be stored in a PL/SQL variable and printed on the screen.

Note: SET VERIFY OFF in the PL/SQL block.

SOLUTION:

```
1  -- Q3
2  ✓ DECLARE
3      p_num1 NUMBER := 2;
4      p_num2 NUMBER := 4;
5      v_result NUMBER;
6  ✓ BEGIN
7      v_result := (p_num1 / p_num2) + p_num2;
8      DBMS_OUTPUT.PUT_LINE('The result is: ' || v_result);
9  END;
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
The result is: 4.5				
Statement processed.				
0.00 seconds				

Q4. Build a PL/SQL block that computes the total compensation for one year.

- a) The annual salary and the annual bonus percentage values are defined using the DEFINE command. (e.g., DEFINE p_salary = 50000 and DEFINE p_bonus = 10)
- b) Pass the values defined in the above step to the PL/SQL block through iSQL*Plus substitution variables. The bonus must be converted from a whole number to a decimal (for example, 15 to .15). If the salary is null, set it to zero before computing the total compensation. Execute the PL/SQL block. If the salary is null, set it to zero before computing the total compensation. Execute the PL/SQL block.

Reminder: Use the NVL function to handle null values.

Note: Total compensation is the sum of the annual salary and the annual bonus. To test the NVL function, set the DEFINE variable equal to NULL.

DEFINE p_salary =50000

DEFINE p_bonus=10

SOLUTION:

```
11  -- Q4
12  DECLARE
13      p_salary      NUMBER := 50000;
14      p_bonus       NUMBER := 10;
15      v_salary      NUMBER;
16      v_bonus_pct   NUMBER;
17      v_total_comp  NUMBER;
18  BEGIN
19      v_bonus_pct := p_bonus / 100;
20      v_salary := NVL(p_salary, 0);
21      v_total_comp := v_salary + (v_salary * v_bonus_pct);
22
23      DBMS_OUTPUT.PUT_LINE('Total Amount: ' || v_total_comp);
24  END;
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
Total Amount: 55000				
Statement processed.				
0.00 seconds				